

Week 2

Pressure, density, and dynamic pressure

Automotive Aerodynamics Independent Study

Industrial Design × Physics (DePaul)

Physics mentor · 10-week IS

Today's agenda

1. Recap Week 1: forces from the CM, $N = mg + D_w$
2. Where aero forces come from: pressure (and a little shear)
3. Units: pascal
4. Why **differences** in P matter, not P_{atm} itself
5. Density ρ : meaning and units
6. Kinetic energy reminder $\rightarrow$ energy per volume $\rightarrow q$
7. Why aero loads grow like v^2
8. Worked checkpoint
9. Pressure-region sketches: wing and splitter
10. Target test speeds and q
11. Deliverables

Learning targets (by end of Week 2)

You will be able to:

- Use $P = F/A$ with unit $\text{Pa} = \text{N}/\text{m}^2$
- Define density $\rho = m/V$ (mass per volume, unit kg/m^3)
- Derive $q = \frac{1}{2}\rho v^2$ as kinetic energy per volume of the stream
- Compute q and explain why aero forces grow roughly with v^2
- Map qualitative high/low pressure regions on a wing or splitter **as hypotheses**
- Choose a target model airspeed and convert it to q

Recap — Week 1 still governs

Particle-model FBD: every force is drawn from the center of mass.

$$N = mg + D_w, \quad F_{\text{drive}} = F_D \quad (a = 0)$$

Downforce does **not** change mass. It changes **load** N , so the tire friction budget $f_{\text{max}} \sim \mu N$ can rise at speed.

Week 2 asks: what in the air produces D_w and F_D ?

Learning outcome LO2 (syllabus)

Use pressure, density, and dynamic pressure $q = \frac{1}{2} \rho v^2$ correctly.

Week 3 will wrap q into C_D and C_L .

Week 4 will add Bernoulli — and when it lies.

Do not skip to slogans. Get P , ρ , and q solid first.

Fluids push; net force is an imbalance

Air is a fluid. It pushes on every exposed face.

Two contributions to the force on a body:

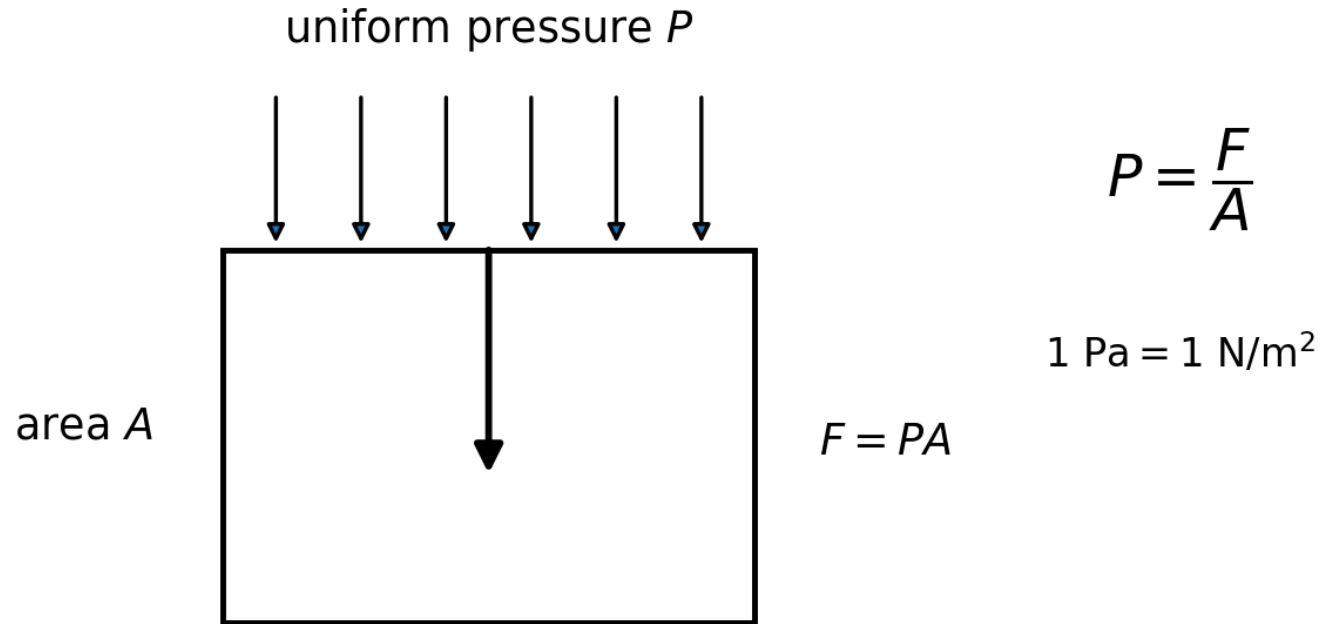
Contribution	Everyday name	Week 2 status
Pressure (normal to the surface)	"push"	Main story
Viscous shear (tangent)	"skin friction"	Mention; measure later as part of F_D

For spoilers, wings, splitters, and bluff bodies, **pressure imbalance** usually dominates the force you care about.

Pressure

$$P = \frac{F}{A}, \quad [P] = \text{Pa} = \text{N/m}^2$$

Pressure is force per area (resultant into the surface)



Atmosphere is huge — and mostly cancels

Standard sea-level pressure:

$$P_{\text{atm}} \approx 1.01 \times 10^5 \text{ Pa} = 101 \text{ kPa}$$

On a 0.02 m^2 plate that would be

$$F_{\text{one side}} \approx (1.01 \times 10^5)(0.02) \approx 2000 \text{ N}$$

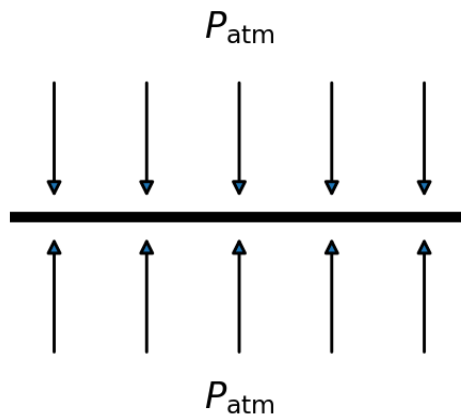
You do **not** feel 2000 N of “air weight” on a notebook. The other side is also at $\sim P_{\text{atm}}$.

Language: aero force is not “air pressure.” It is ΔP .

Why differences?

Aero force comes from pressure differences

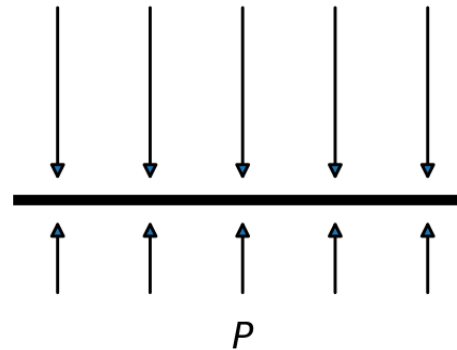
Same P both faces



net $F = 0$

Difference ΔP

$P + \Delta P$



net $F \approx \Delta P A$ (down)

Why differences? (net force)

If the two faces of a thin part see P and $P + \Delta P$,

$$F_{\text{net}} \approx \Delta P A$$

Direction: from the higher- P face toward the lower- P face.

Checkpoint preview — Problem 3

Why is “air pressure” alone not enough?

Because P_{atm} on opposite faces nearly cancels.

The force that shows up on a load cell is about the **imbalance** ΔP , plus shear.

If your CAD caption says “high pressure underneath,” you must mean **higher than the other side**, not “the air has pressure.”

Density ρ — what the symbol means

$$\rho \equiv \frac{m}{V}$$

Read it as **how much mass is packed into a volume**.

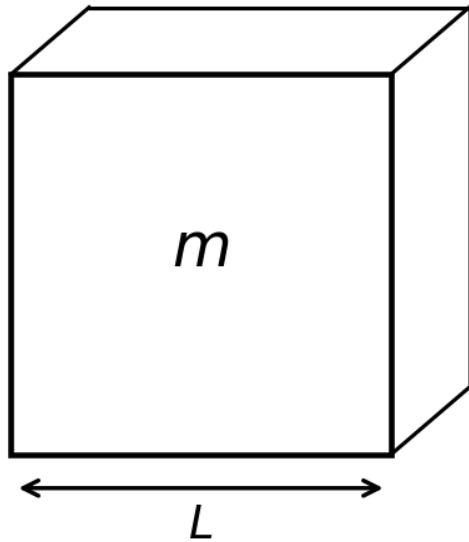
For a cube of side L , $V = L^3$.

SI unit: kg/m^3 (kilograms of stuff per cubic meter of space).

Air is not “nothing”: a cubic meter of room air still has about a kilogram of mass.

Density as a picture

Density is mass per volume — how much air is in a box



$$\text{density } \rho = \frac{m}{V}$$

$$V = L^3$$

$$[\rho] = \text{kg/m}^3$$

$$\text{air (room): } \rho \approx 1.2 \text{ kg/m}^3$$

Working value for this course

Near room conditions

$$\rho_{\text{air}} \approx 1.2 \text{ kg/m}^3$$

Use this unless you measure better (temperature, altitude, humidity).

This term: freeze $\rho = 1.2 \text{ kg/m}^3$ in early calculations; note it in the lab notebook if the weather changes.

Denser air (cold day, sea level) means **more mass in the same volume**, so the same v carries more kinetic energy — and q is larger.

Reminder: kinetic energy (intro physics)

A particle of mass m moving at speed v has

$$KE = \frac{1}{2}mv^2$$

Units: joules, $J = N \cdot m = kg \cdot m^2/s^2$.

Air is many such particles. We do not track each one. We track a **blob** of air with total mass m and volume V .

That blob still has $KE = \frac{1}{2}mv^2$ if the blob moves at v with the free stream.

Energy per volume is a pressure

Divide the blob's kinetic energy by its volume:

$$\frac{KE}{V} = \frac{1}{2} \left(\frac{m}{V} \right) v^2 = \frac{1}{2} \rho v^2$$

That is why ρ appears: m/V is ρ .

That is why v^2 appears: it was already in KE .

Units check (this is the point of the algebra):

$$\frac{\text{J}}{\text{m}^3} = \frac{\text{N} \cdot \text{m}}{\text{m}^3} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$$

Energy per volume has the same units as pressure. We give that scale a name.

From kinetic energy to q (summary)

From intro-physics KE to q (why ρ and v^2 appear)

kinetic energy of a blob of air: $KE = \frac{1}{2}mv^2$

divide by the blob's volume V :

$$\frac{KE}{V} = \frac{1}{2}\left(\frac{m}{V}\right)v^2 = \frac{1}{2}\rho v^2$$

that energy density is dynamic pressure $q \equiv \frac{1}{2}\rho v^2$

units: $\text{J/m}^3 = \text{N} \cdot \text{m/m}^3 = \text{N/m}^2 = \text{Pa}$

Dynamic pressure

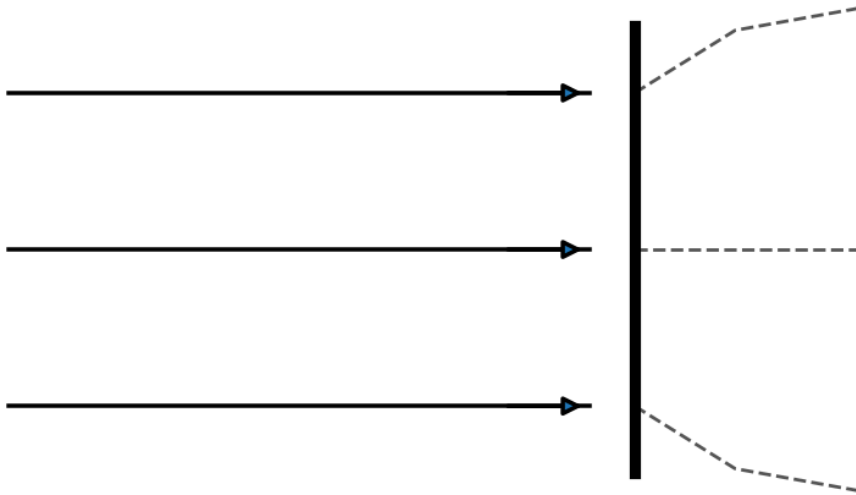
$$q \equiv \frac{1}{2} \rho v^2$$

Same units as pressure (Pa): characteristic inertial ("ram") scale of the free stream.

Oncoming air carries q ; the solid turns that into a force

free stream v (stops / splits at the solid)

plate, area A



$$q = \frac{1}{2} \rho v^2$$

inertial pressure scale
of the oncoming air (Pa)

$$F \sim qA \text{ before } C_D, C_L$$

Why v^2 ?

Because $KE \propto v^2$, the energy per volume $q = \frac{1}{2}\rho v^2$ also $\propto v^2$.

Doubling speed **quadruples** that scale. ρ is the conversion from “a mass m ” to “a fluid filling volume V .”

So if ΔP is some fraction of q , aero **forces** also grow like v^2 (until the flow pattern itself changes — stall, Week 7; Reynolds number, Week 5).

This is why:

- Race-car aero “comes on” with speed
- A small model at modest fan speed may still produce only a few newtons
- Safety: loads explode if you crank the fan thoughtlessly

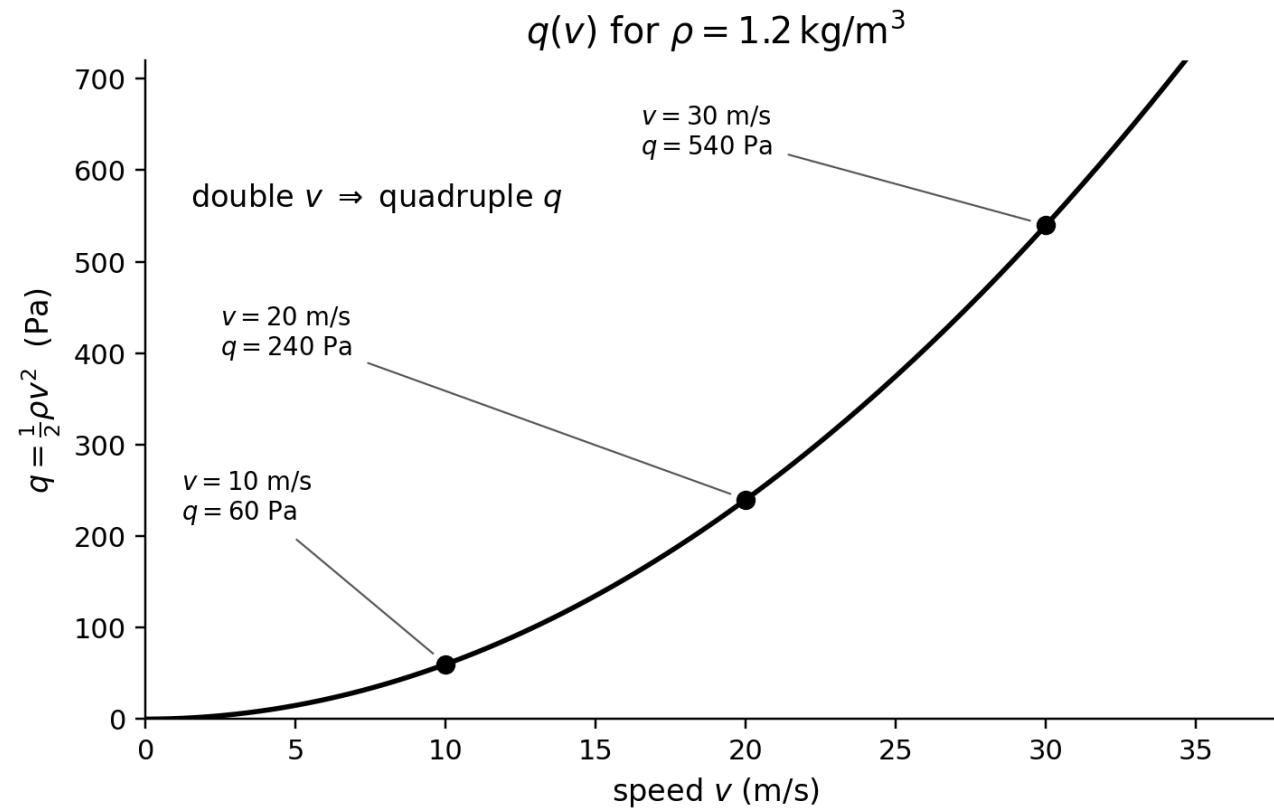
Worked q (checkpoint 1)

Use $\rho = 1.2 \text{ kg/m}^3$.

$v \text{ (m/s)}$	v^2	$q = \frac{1}{2}\rho v^2 \text{ (Pa)}$
10	100	$0.6 \times 100 = 60$
20	400	$0.6 \times 400 = 240$
30	900	$0.6 \times 900 = 540$

$20 = 2 \times 10$ but $240 = 4 \times 60$. **Factor of 4 when speed doubles.**

The $q(v)$ curve



Mark your planned fan / tunnel speeds on a copy of this plot.

Order of magnitude: $F \sim qA$

Before coefficients (Week 3), estimate

$$F \sim qA$$

Example from the notes: $v = 20 \text{ m/s}$, $q = 240 \text{ Pa}$, $A = 0.020 \text{ m}^2$

$$qA = 4.8 \text{ N}$$

If a sensor cannot resolve $\sim 0.1 \text{ N}$, this speed/area combination may still be usable. If qA is 0.2 N , worry (Week 1 scale-model note).

Checkpoint 2 — $\Delta P A$

| $A = 0.020 \text{ m}^2$, average $\Delta P = 150 \text{ Pa}$. Force magnitude? Direction?

$$F \approx \Delta P A = (150)(0.020) = 3.0 \text{ N}$$

Direction: high $P \rightarrow$ low P .

You must mark which face is which on the sketch. The number alone has no meaning without that.

Compare: q at 20 m/s is 240 Pa , so $\Delta P = 150 \text{ Pa}$ is a plausible fraction of q ($C \sim \Delta P/q \sim 0.6$).

Design sketches are hypotheses

You will annotate CAD: "this face high P ," "that face low P ."

Those marks are **predictions**, not decorations. Week 6–9 sensors test them.

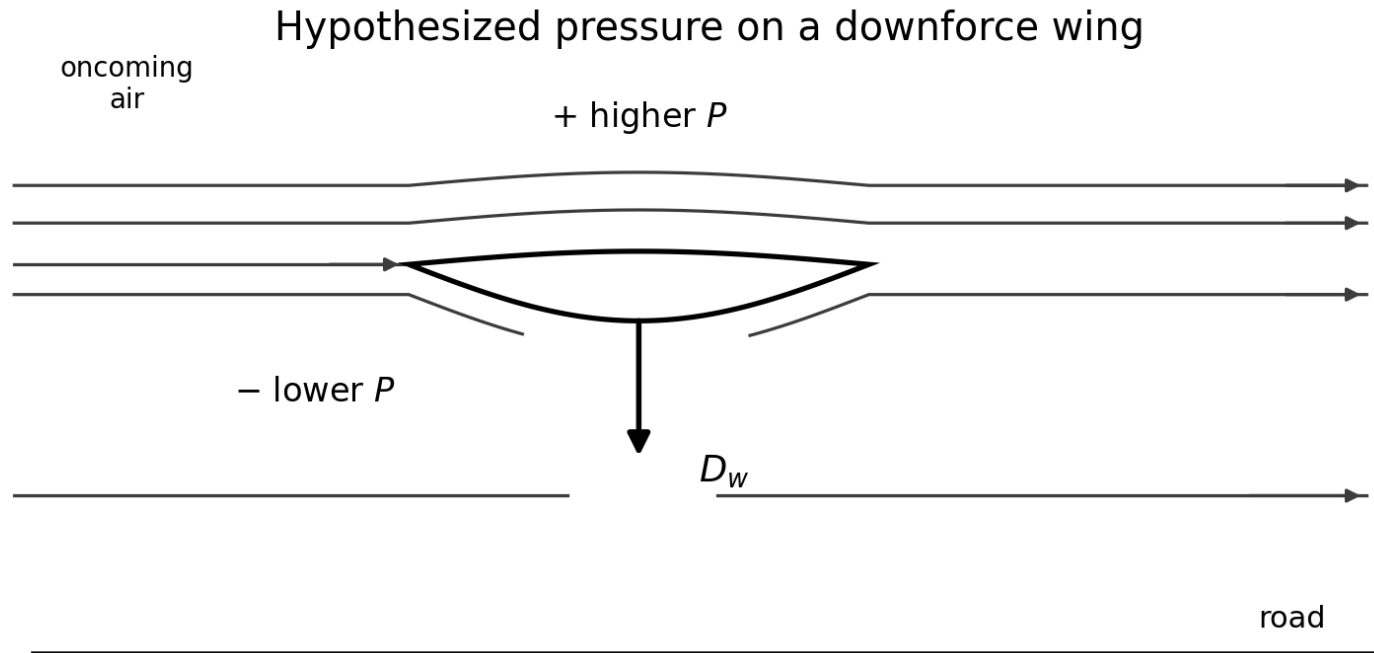
Week 4 will explain Bernoulli's $P + \frac{1}{2}\rho v^2$ along a streamline — and why a stalled spoiler is **not** a Bernoulli story.

Today: qualitative maps only, tied to ΔP and q .

Inverted wing (downforce)

Airplane wing: lower P on top $\Rightarrow$ lift **up**.

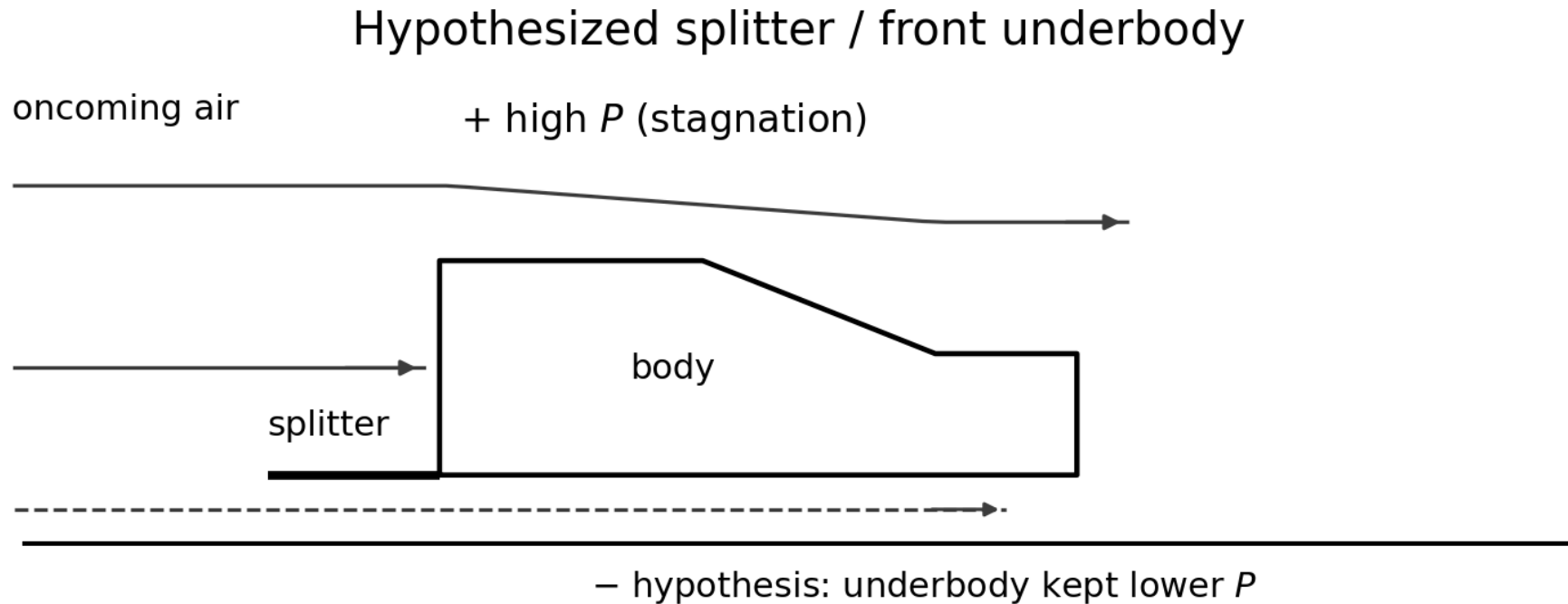
Car wing: inverted $\Rightarrow$ lower P toward the road $\Rightarrow$ net force **down**.



Streamlines split around the solid. They do not wrap through it.
Net pressure force on this inverted wing is into the road.

Splitter (front)

Hypothesis: the splitter sees high pressure near the nose (stagnation) and helps keep that high pressure from flooding the underbody, so the underbody can run lower P .



Hypothesis only. Streamlines do not pass through the body. Week 4: Bernoulli limits.

In-class: annotate two of *your* parts

Take WING-xx and SPLIT-xx (or SPOIL / DIFF).

For each:

1. Arrow for relative wind
2. + / — on faces
3. Expected net D_w and F_D directions (Week 1 FBD)
4. One sentence: "I will believe this if the load cell shows ..."

These sketches are a Week 2 deliverable.

Target test speed $\rightarrow q$ table

Pick a realistic model speed range (fan, not a highway). Example:

Setting	v (m/s)	q (Pa) at $\rho = 1.2$	qA if $A = 0.015 \text{ m}^2$
low	8	38	0.58 N
mid	12	86	1.3 N
high	16	154	2.3 N

If you cannot measure v yet, record **fan setting** and plan a calibration (Week 6). Still compute q for the v you *hope* to hit.

Physics checkpoint (full)

1. $q = 60,240,540 \text{ Pa}$; doubling v multiplies q by 4.
2. $F \approx 3.0 \text{ N}$, high P toward low P .
3. Net force needs ΔP , because P_{atm} cancels on opposite faces.

Bring units. "About 200" with no Pa is not done.

Design / build tasks this week

- [] Annotated pressure sketches for ≥ 2 parts
- [] Target speed / q table for the platform
- [] If v is unknown: write how you will measure or calibrate it
- [] Keep part IDs from Week 1 (WING-01 , ...)

Physics before filament: a pretty wing with no $+/-$ map is not yet a Week 2 artifact.

Deliverables checklist

Deliverable	Where
Checkpoint 1–3 with units	notes or curriculum/week-02-checkpoint.md
Pressure sketches (≥ 2 parts)	cad/notes/ or docs/
Speed / q table	brief addendum or lab notebook

Due before Week 3 meeting.

How you will be assessed on Week 2

Strong: correct q arithmetic; ΔP language; honest sketches labeled as hypotheses.

Weak: “the air is fast so pressure is low” with no ΔP , no q , and no units.

Save the Bernoulli fight for Week 4 — but do not invent it early as a magic spell.

Looking ahead — Week 3

The engineering models:

$$F_D = q C_D A, \quad F_L = q C_L A$$

You will pick a **reference area** A for the whole term and treat downforce as **negative lift**.

Bring your q table and a first estimate of frontal area.

Instructor talking points

- If v is unknown, do not let q remain decorative — force a measurement plan.
- Watch for “the atmosphere pushes the car down.” Redirect to ΔP .
- Watch for ρ as “how thick the air looks.” It is **mass per volume**.
- Keep Bernoulli qualitative at most; Week 4 owns the assumptions.
- Connect every pressure sketch back to the Week 1 FBD (where does D_w sit?).

References for this lecture

- Course note: [physics/02-pressure-and-dynamic-pressure.md](#)
- Week plan: [curriculum/week-02-pressure-and-dynamic-pressure.md](#)
- Week 1 FBD: [physics/01-forces-and-fbds.md](#)
- Syllabus LO2: [curriculum/SYLLABUS.md](#)
- Figures: [curriculum/slides/figures/](#)

End of Week 2 lecture

Next actions

1. Finish checkpoint with units
2. Annotate two parts
3. Publish a $v-q$ table
4. Push to the course GitHub repo

Questions?